
MULTI-TRACE QUARTIC MATRIX MODEL USING MARKOV CHAIN MONTE CARLO

PROJECT REPORT



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July 12, 2019

Abstract

In this project we simulate Quartic Matrix Models with and without interactions using the method of Markov Chain Monte Carlo (MCMC). We discuss Monte Carlo method of sampling and its usefulness in numerical computations. We discuss how random variables can be generated from any complicated probability distribution using the MCMC method. We simulate quantum harmonic oscillator using MCMC. In this case, we know the exact solutions and we use them to cross-check our simulation results. We also simulate Real Quartic Matrix Model using MCMC and study the eigenvalue distributions in this model. We observe a phase transition in the pure matrix model. We also study the interacting case, and, there we observe that the width of the eigenvalue distribution narrows as the interaction strength is increased.

Acknowledgements

I would like to show my gratitude to Dr. Anosh Joseph, Project Supervisor at IISER Mohali for giving me good guidelines throughout the project. My special thanks to my friend Mr. Adhrit Ravichandran who helped me in every difficulties related to C++ programming I faced.

I would like to thank the institute, IISER Mohali, for giving this opportunity to visit here. I would like to thank everyone involved in organizing this Summer Program at IISER Mohali and making my stay here smooth. I would like to thank all my professors and my institute, IISER Kolkata, for making capable of standing here.

Last but not the least, I would like to expand my thanks to everyone who directly or indirectly helped me in completing this project.

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IISER Kolkata
July 2019

Contents

1	Monte Carlo Method of Sampling	iii
1.1	General Idea of the Method	iii
1.2	Uses of the Method	iii
1.2.1	Computing Integrals	iii
1.2.2	Probabilities	iv
1.2.3	Volume of any Set	iv
1.3	Importance Sampling	iv
2	Markov Chain monte Carlo (MCMC)	vi
2.1	Markov Chain	vi
2.2	Metropolis Algorithm	vi
2.3	Advantages of of MCMC	vii
3	Quantum Harmonic Oscillator	viii
3.1	The Model	viii
3.2	Discretization	ix
4	Real Quartic Matrix Model	xi
4.1	The Pure Real Quartic Matrix Model	xi
4.2	The Multi-Trace Quartic Matrix Model	xi
4.3	The Eigenvalue Distribution	xii
4.4	Metropolis Algorithm	xii
4.5	Plots of Distribution of Eigenvalues	xiv
5	Conclusions	xviii
	Bibliography	xix

List of Figures

1.1	The idea of importance sampling. Here $f(x)$ is the function whose expectation value we are calculating with respect to $p(x)$, and $q(x)$ is the proposal function for importance sampling.	iv
3.1	Acceptance rate against Monte Carlo time for different values of the simulation parameter Δ	ix
3.2	Correlation function $\langle x(\tau)x(0) \rangle$ against Euclidean time τ	x
4.1	The Monte Carlo time history of one of the eigenvalues λ of the pure quartic matrix model. We see that it thermalizes after about 300 steps. . .	xiii
4.2	The autocorrelation against the lag time T . The solid curve is a fit to the data.	xiv
4.3	Eigenvalue distributions for different values of coupling parameter g with interaction strength $D = 0$	xv
4.4	Combined eigenvalue distributions for coupling parameter $g = 0.5, 0.8, 2, 3$ with interaction strength $D = 0$	xvi
4.5	Eigenvalue distributions for different values of interaction strength D with fixed coupling parameter $g = 2.0$	xvi
4.6	Combined eigenvalue distributions for $D = 0.4, 0.8, 2, 8$ with $g = 2.0$	xvii

1 Monte Carlo Method of Sampling

Let $\vec{X} = (X_1, X_2, X_3, \dots, X_n)$ be a n -dimensional real random variable, i.e., $\vec{X} \in \mathbb{R}^n$ with the probability distribution function (PDF) $p(\vec{X})$. Our goal is to calculate the expectation value of any real valued function $f(\vec{X})$ with respect to this probability distribution, defined by

$$E_p(f) := \int_A f(\vec{X}) p(\vec{X}) d^n X, \quad (1.1)$$

where $A \subset \mathbb{R}^n$ is the domain of the random variables. If we can generate random vectors $\vec{X}$ which follows the PDF $p(\vec{X})$ then Monte Carlo method gives a way of approximating this expectation value.

1.1 General Idea of the Method

If $\vec{X}_1, \vec{X}_2, \vec{X}_3, \dots, \vec{X}_k \sim p$ be identically and independently distributed (iid) random vectors, then we define the k -th Monte Carlo estimator to be

$$\mu_k := \sum_{i=1}^k f(\vec{X}_i). \quad (1.2)$$

Then the Monte Carlo method approximates the expectation value by the k -th Monte Carlo estimator. Higher value of k means that we are closer to the exact value.

1.2 Uses of the Method

1.2.1 Computing Integrals

We can calculate integrals of any functions by converting it to an expectation value as in

$$\begin{aligned} \int_A f(\vec{X}) d^n X &= m(A) \int_A f(\vec{X}) \frac{1_A}{m(A)} d^n X \\ &= m(A) E_{U_A}(f), \end{aligned} \quad (1.3)$$

where

$$U_A = \frac{\mathbb{1}_A}{m(A)}, \quad (1.4)$$

is the uniform PDF on A and $m(A)$ is the volume of A .

1.2.2 Probabilities

Probability of finding $\vec{X}$ in $A \subset \mathbb{R}$ is given by

$$Prob_p(A) = \int_A p(\vec{X}) d^n X, \quad (1.5)$$

which can be expressed as $E_p(\mathbb{1}_A)$ or $m(A)E_{U_A}(p)$.

1.2.3 Volume of any Set

The volume $m(A)$ of a bounded subset A can be approximated by $m(B)Prob_{U_B}(A)$, where B is a rectangle containing A , whose volume $m(B)$ we know exactly.

1.3 Importance Sampling

Suppose we want to sample x from a probability distribution $p(x)$ and calculate the expectation value $E_p(f)$ as shown in Fig. 1.1

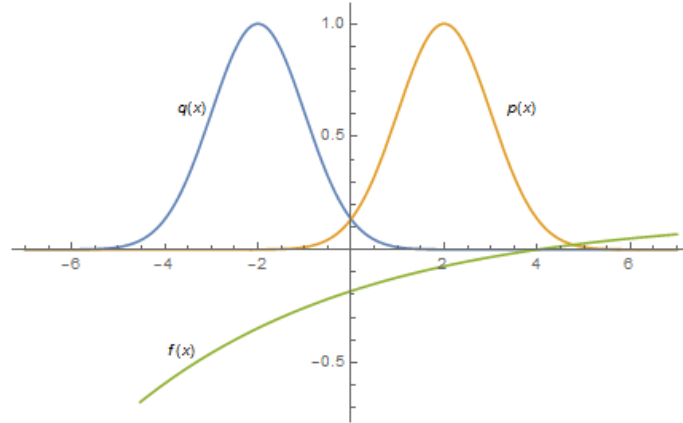


Figure 1.1: The idea of importance sampling. Here $f(x)$ is the function whose expectation value we are calculating with respect to $p(x)$, and $q(x)$ is the proposal function for importance sampling.

1 Monte Carlo Method of Sampling

If we sample x directly from the PDF $p(x)$, then x will be sampled from the portion of the graph where $f(x)$ has less contributions. We can solve this problem by writing the expectation value as

$$\begin{aligned} E_p(f) &= \int f(x)p(x)dx \\ &= \int f(x)\frac{p(x)}{q(x)}q(x)dx \\ &= E_q\left(f(x)\frac{p(x)}{q(x)}\right), \end{aligned} \tag{1.6}$$

where we have introduced a new PDF $q(x)$ whose contribution is very close to $f(x)$ in the required range. This provides a better and faster method to approximate the expectation value.

2 Markov Chain monte Carlo (MCMC)

Suppose the probability distribution we want to sample from is itself very complicated and whose closed form of the cumulative distribution function (CDF) is not known. Using inverse method to sample from the distribution might not be computationally feasible in such cases or for some other reasons. MCMC method provides a computationally faster method of sampling in such cases.

2.1 Markov Chain

Markov chain is a sequence of random variables where the elements of the sequence can be thought of as different states of a variable at different discrete times, and the state of the variable at any time depends only on the state of the variable attained at the previous time.

We say $\{X_n\}_{n \geq 0}$ is a Markov Chain with the initial distribution $\pi(x)$ and transition matrix T if for all $n \geq 0$

- (i) $P(X_0 = a) = \pi(a)$,
- (ii) $P(X_{n+1} = a | X_0 = b_1, \dots, X_n = b_n) = P(X_{n+1} = a | X_n = b_n) = T(a, b_n)$.

2.2 Metropolis Algorithm

Suppose we want to sample from a probability mass function (pmf) π which is known up to an unknown (difficult to calculate) normalization constant Z

$$\pi = \frac{\tilde{\pi}}{Z}. \quad (2.1)$$

We need to choose a symmetric matrix $Q(a, b)$, which we will call the proposal matrix for this algorithm. We can construct a Markov Chain $\{X_n\}_{n \geq 0}$, which converges the probability distribution π . The algorithm for this is given as follows

- (i) Fix the initial state $X_0 = a$ to be in any state a .
- (ii) At i^{th} step, sample x from $Q(x, X_i)$. Sample u from a uniform distribution on $(0,1)$: $U(0, 1)$.

2 Markov Chain monte Carlo (MCMC)

- (iii) If $u < \frac{\tilde{\pi}(x)}{\tilde{\pi}(X_i)}$, then assign $X_{i+1} = x$ else $X_{i+1} = X_i$.

This creates a Markov Chain and it can be shown that independent of the initial state this converges to the probability distribution π .

Thus at each step, the state is updated with an acceptance probability

$$\min \left(\frac{\tilde{\pi}(X_i)}{\tilde{\pi}(X_i)}, 1 \right). \quad (2.2)$$

2.3 Advantages of of MCMC

- (i) No need to compute the CDF of the distribution.
- (ii) No need to know the normalization constant.

3 Quantum Harmonic Oscillator

Here we will calculate the expectation values of some observables in the ground state of a quantum harmonic oscillator.

3.1 The Model

The action for a simple harmonic oscillator is given by

$$S[x(t)] = \int_{t_i}^{t_f} dt \left(\frac{1}{2} m \dot{x}^2 - \frac{1}{2} m \omega^2 x^2 \right), \quad (3.1)$$

which after wick rotation takes the form

$$S[x(\tau)] = \int_{\tau_i}^{\tau_f} d\tau \left(\frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \omega^2 x^2 \right). \quad (3.2)$$

The ground state energy expectation value of $\mathcal{T}\{x(\tau_2)x(\tau_1)\}$ is given by

$$\langle 0 | \mathcal{T}\{x(\tau_2)x(\tau_1)\} | 0 \rangle = \frac{\int [\mathcal{D}x] x(\tau_2) x(\tau_1) e^{-\frac{1}{\hbar} \int_{-\infty}^{\infty} d\tau \left(\frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \omega^2 x^2 \right)}}{\int [\mathcal{D}x] e^{-\frac{1}{\hbar} \int_{-\infty}^{\infty} d\tau \left(\frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \omega^2 x^2 \right)}}. \quad (3.3)$$

Thus

$$\pi[x(\tau)] = \frac{e^{-\frac{1}{\hbar} \int_{-\infty}^{\infty} d\tau \left(\frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \omega^2 x^2 \right)}}{Z}, \quad (3.4)$$

can be treated as a probability distribution over the space of paths in the ground state where the partition function Z is

$$Z = \int [\mathcal{D}x] e^{-\frac{1}{\hbar} \int_{-\infty}^{\infty} d\tau \left(\frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \omega^2 x^2 \right)}. \quad (3.5)$$

MCMC method can be used to sample from this distribution after discretizing the Euclidean time direction τ .

3.2 Discretization

For discretization to work correctly we need to put periodic boundary condition by identifying the initial and final times: $\tau_i \sim \tau_f$. After this, we discretize it with N sites of length a , so that $\tau_f - \tau_i = Na$.

After discretization, the change in action corresponding to a random shift in x_i , i.e.,

$$x_i \rightarrow x_i + \delta, \quad (3.6)$$

where δ is chosen uniformly in the range

$$-\Delta \leq \delta \leq \Delta \quad (3.7)$$

is

$$\frac{\delta S}{\hbar} = \delta [\delta + 2x_i - g(x_{i-1} + x_{i+1})]. \quad (3.8)$$

This shift in x_i is accepted with a probability of

$$\min\left(1, e^{-\frac{\delta S}{\hbar}}\right). \quad (3.9)$$

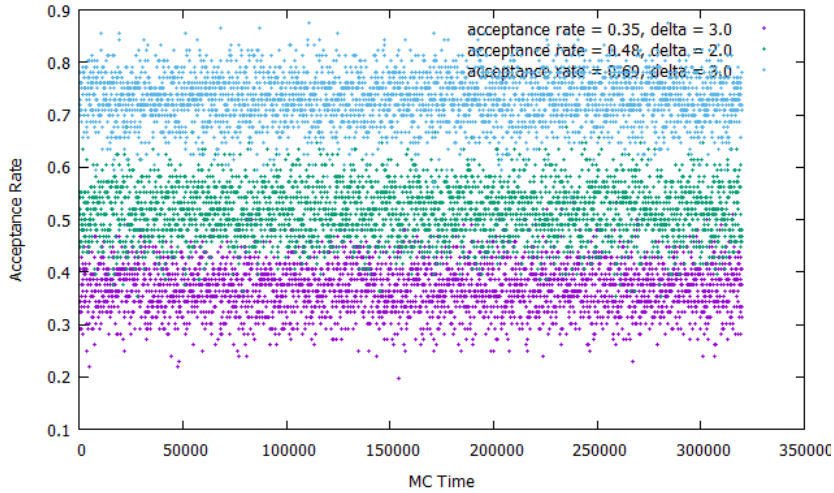


Figure 3.1: Acceptance rate against Monte Carlo time for different values of the simulation parameter Δ .

In Fig. 3.1 we show the average acceptance rate for different values of maximum shift Δ at each Monte Carlo time step. We see that the acceptance rate is inversely proportional to Δ .

In Fig. 3.2 we show the correlation function $\langle x(\tau)x(0) \rangle$ against the Euclidean time τ . It is fitted with the exact result $\langle x(\tau)x(0) \rangle = A \cosh(\omega(\tau - N/2))$. It fits very well and this shows the validity of our sampling method.

3 Quantum Harmonic Oscillator

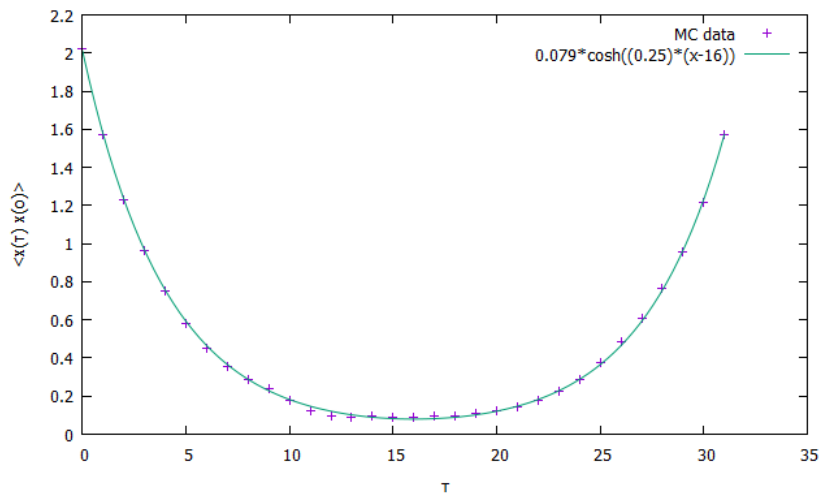


Figure 3.2: Correlation function $\langle x(\tau)x(0) \rangle$ against Euclidean time τ .

4 Real Quartic Matrix Model

These are models on the space of Hermitian Matrices (considered as a vector space over the real field).

4.1 The Pure Real Quartic Matrix Model

This is a model describing a single hermitian matrix without any interaction. This is given by the action

$$\begin{aligned} S &= B \operatorname{Tr} M^2 + C \operatorname{Tr} M^4 \\ &= \frac{N}{g} \left(-\operatorname{Tr} M^2 + \frac{1}{4} \operatorname{Tr} M^4 \right), \end{aligned} \quad (4.1)$$

where N is the number of rows of the matrix and g is a coupling parameter.

The partition function is given by

$$Z = \int dM e^{-S}, \quad (4.2)$$

here dM is a measure on the space of hermitian matrices

$$dM = \prod_{i=1}^N dx_{ii} \prod_{i < j} dx_{ij} dy_{ij}, \quad (4.3)$$

where $x_{ij} = \Re(M_{ij})$ and $y_{ij} = \Im(M_{ij})$, $1 \leq i < j \leq N$ are respectively, the real and imaginary parts of the ij^{th} element.

4.2 The Multi-Trace Quartic Matrix Model

In this model an interaction term is added to the above model

$$S = \operatorname{Tr} (BM^2 + CM^4) + D (\operatorname{Tr} M^2)^2. \quad (4.4)$$

4.3 The Eigenvalue Distribution

The measure dM can be written as

$$dM = d\Lambda dU \Delta^2(\Lambda), \quad (4.5)$$

where $d\Lambda$ is the measure over the eigenvalues

$$d\Lambda = \prod_{i=1}^N d\lambda_i, \quad (4.6)$$

dU is the Haar measure over the group $SU(N)$ with normalization $\int dU = 1$, and $\Delta^2(\Lambda)$ is the Vandermonde determinant

$$\Delta^2(\Lambda) = \prod_{1 \leq i < j \leq N} (\lambda_i - \lambda_j)^2. \quad (4.7)$$

Integrating with respect to the Haar measure the partition function for the Multi-Trace Quartic Matrix Model becomes

$$\begin{aligned} Z &= \int d\Lambda \Delta^2(\Lambda) \exp \left(-\text{Tr} (B\Lambda^2 + C\Lambda^4) - D (\text{Tr} \Lambda^2)^2 \right) \\ &= \int d\Lambda \exp \left(B \sum_{i=1}^{\infty} \lambda_i^2 + C \sum_{i=1}^{\infty} \lambda_i^4 + D \left(\sum_{i=1}^{\infty} \lambda_i^2 \right)^2 - \frac{1}{2} \sum_{i \neq j} \ln (\lambda_i - \lambda_j)^2 \right) \\ &= \int d\Lambda \exp (S_{eff}). \end{aligned} \quad (4.8)$$

This effective action now describes a model of the space of N -tuples of N eigenvalues. This gives the distribution of the eigenvalues in the model.

4.4 Metropolis Algorithm

We will use the Metropolis algorithm to study this model.

Under the change

$$\lambda_i \longrightarrow \lambda_i + h, \quad (4.9)$$

with h being a random number chosen from a uniform distribution $[-0.5, 0.5]$, the effective potential S_{eff} is modified in the following way

$$S_{eff} \longrightarrow S_{eff} + \Delta S_{i,h}, \quad (4.10)$$

where

$$\Delta S_{i,h} = B\Delta S_2 + C\Delta S_4 + D (2S_2\Delta S_2 + \Delta S_2^2) + \Delta S_{\text{Vand}}, \quad (4.11)$$

4 Real Quartic Matrix Model

with

$$\Delta S_2 = h^2 + 2h\lambda_i, \quad (4.12)$$

$$\Delta S_4 = 6h^2\lambda_i^2 + 4h\lambda_i^3 + 4h^3\lambda_i + h^4, \quad (4.13)$$

$$\Delta S_{\text{Vand}} = -2 \sum_{j \neq i} \ln \left| 1 + \frac{h}{\lambda_i - \lambda_j} \right|. \quad (4.14)$$

We have used a uniform distribution on $[-0.5, 0.5]$ as the proposal distribution to choose h at each Monte Carlo time step. This choice gave us an average acceptance rate of about 60% during the Metropolis updates.

In Fig. 4.1 we show thermalization of the eigenvalues of our model with rank 5 matrices. We chose -40 as the initial values for the eigenvalues.

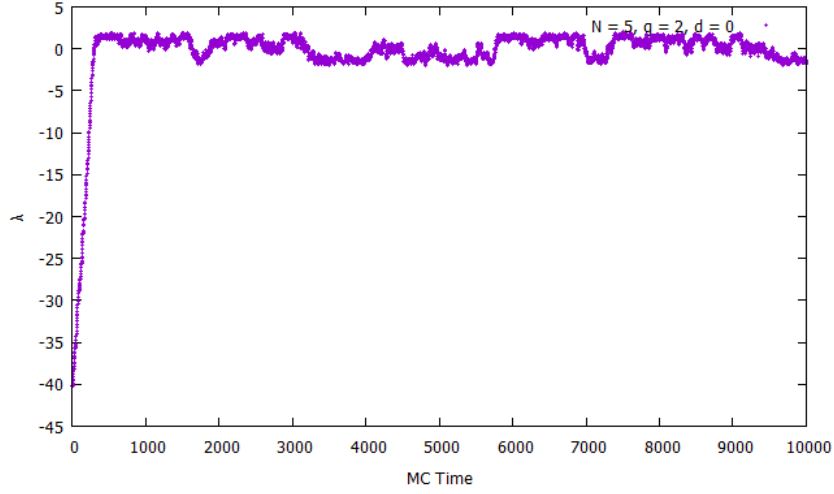


Figure 4.1: The Monte Carlo time history of one of the eigenvalues λ of the pure quartic matrix model. We see that it thermalizes after about 300 steps.

In Fig. 4.2 we show the autocorrelation with respect to the lag time. It is fitted with the function

$$f(T) = Ae^{-\frac{T}{\tau}}, \quad (4.15)$$

where T is the lag time and τ is called the autocorrelation time. After the fit, the autocorrelation time is coming out to be $\tau = 629$. This shows that the eigenvalues must be generated with a lag time much higher than 629 to get reliable results.

4 Real Quartic Matrix Model

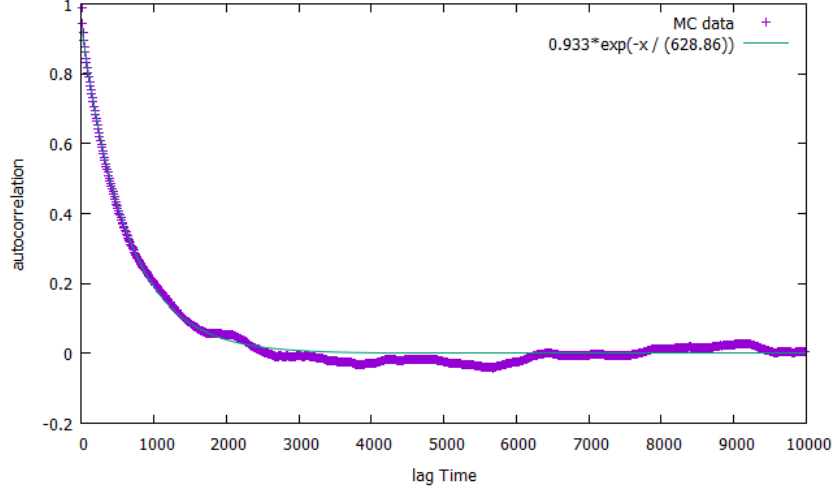


Figure 4.2: The autocorrelation against the lag time T . The solid curve is a fit to the data.

4.5 Plots of Distribution of Eigenvalues

We performed Monte Carlo simulations with Metropolis update for the Pure Quartic Matrix Model (with no interactions). In Fig. 4.3 we show the eigenvalue distributions for various values of $g = 0.3, 0.5, 0.8, 1, 2, 3$, with interaction strength $D = 0$. We discarded the first 10,000 Monte Carlo steps as thermalization steps. The data is generated with a lag time of 750.

It is theoretically predicted that there is a phase transition at $g = g_c = 1.0$. We can see that such a transition happens by looking at the plots from our simulations. Values of λ near 0 has very less probability of occurrence when $g = 0.2$ or 0.5 but it increases very rapidly after $g = 1.0$.

In Fig. 4.4 we show the combined eigenvalue distributions for $g = 0.5, 0.8, 2, 3$ for the case $D = 0$.

In Fig. 4.5 we show the probability distributions of eigenvalues for different values of D , the coefficient of the interaction term $(Tr M^2)^2$, keeping the value of g fixed at $g = 2.0$.

In Fig. 4.6 we show the combined eigenvalue distributions for $D = 0.4, 0.8, 2, 8$ with fixed $g = 2.0$. We can see from the plots that as D increases the probability of eigenvalues near zero increases and the plot becomes narrower.

4 Real Quartic Matrix Model

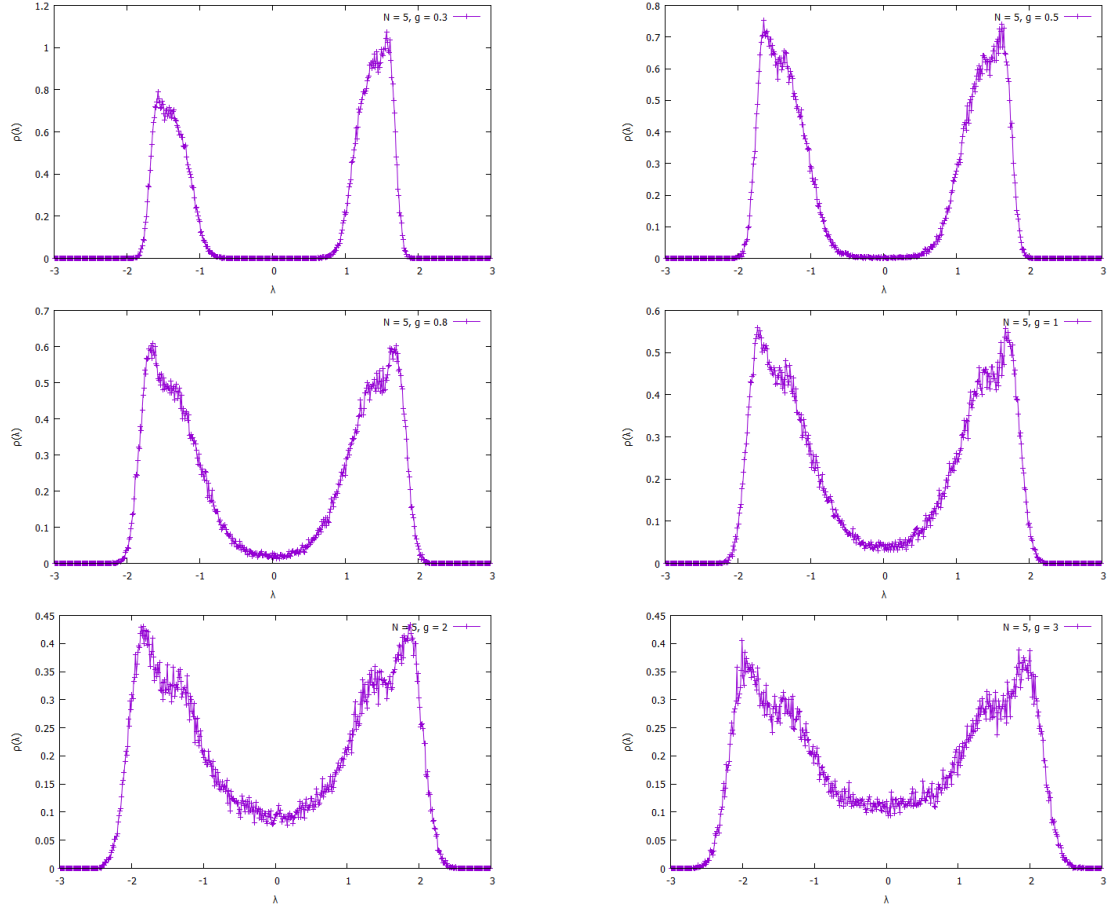


Figure 4.3: Eigenvalue distributions for different values of coupling parameter g with interaction strength $D = 0$.

4 Real Quartic Matrix Model

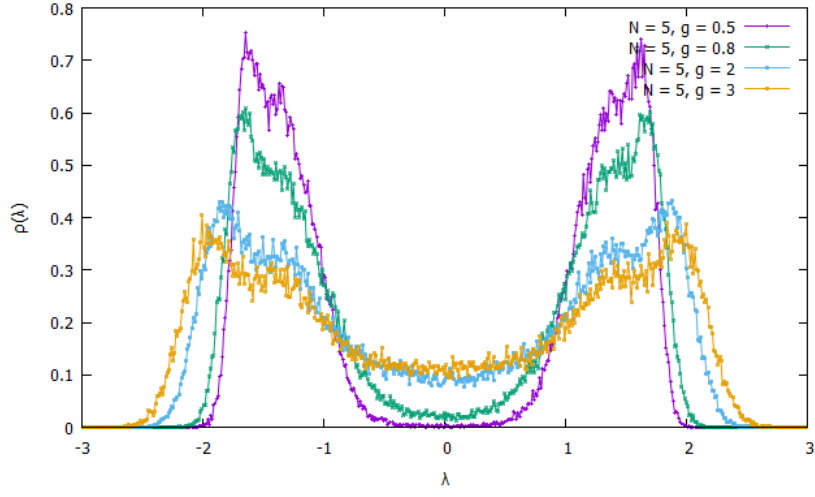


Figure 4.4: Combined eigenvalue distributions for coupling parameter $g = 0.5, 0.8, 2, 3$ with interaction strength $D = 0$.

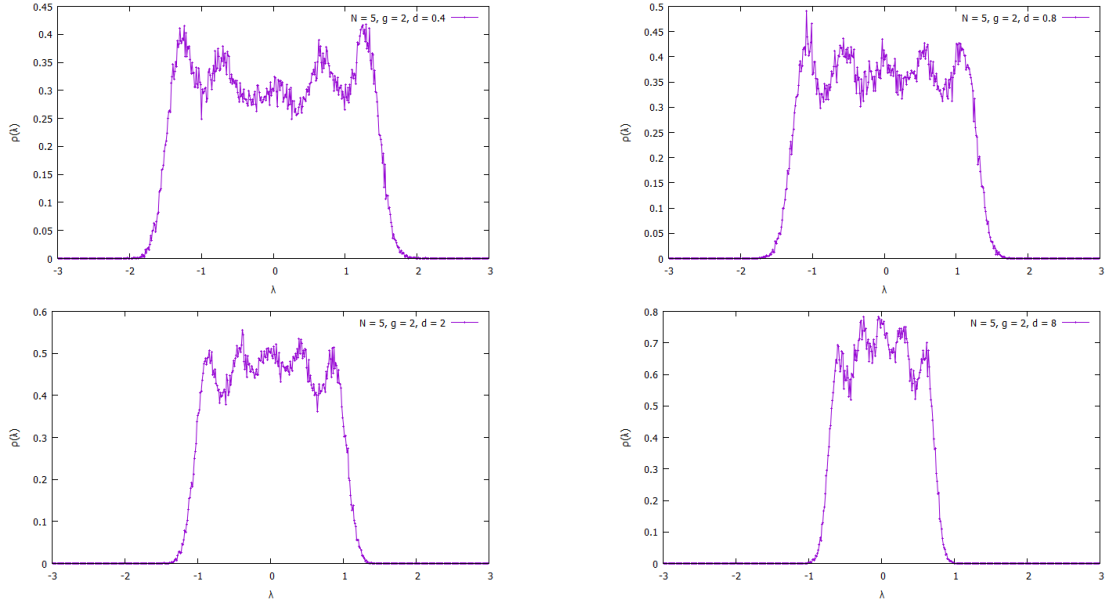


Figure 4.5: Eigenvalue distributions for different values of interaction strength D with fixed coupling parameter $g = 2.0$.

4 Real Quartic Matrix Model

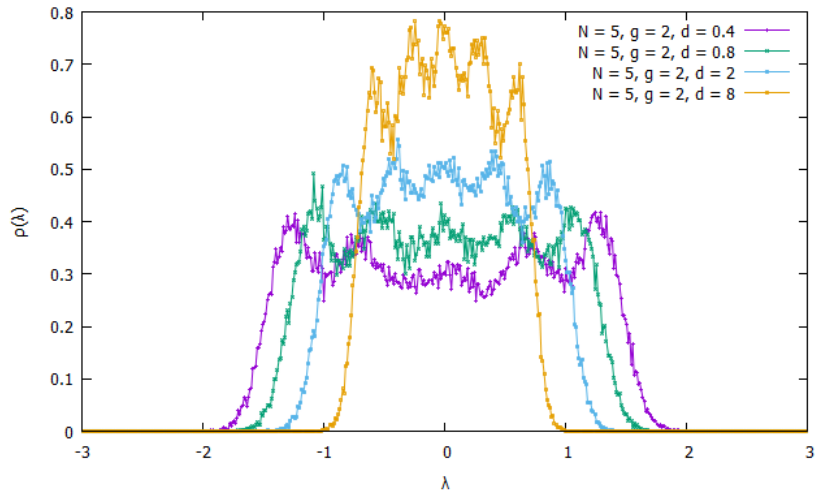


Figure 4.6: Combined eigenvalue distributions for interaction strength $D = 0.4, 0.8, 2, 8$ with fixed coupling parameter $g = 2.0$.

5 Conclusions

We have simulated the Pure and Multi-Trace Quartic Matrix Model using Markov Chain Monte Carlo with Metropolis algorithm. We looked at the eigenvalue distributions for various values of the coupling parameter. We found a phase transition in the Pure Quartic Matrix Model for the value of the coupling $g = 1.0$. In the Multi-Trace Quartic Matrix Model we found that the width of the eigenvalue distributions becomes narrower as the interaction strength D increases.

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